

AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

DEPARTMENT OF CSE

Sessional Test

Program: B.Tech
Session: 2025-26
Subject: FAFL
Max. Marks: 50

Semester: IV
Section: CSE, CS, CSE-DS, CSE-AI/ML, AI/ML, IT, CSIT
Subject Code: BCS-402
Time: 2 Hours

OBE Remarks:

Q.No	1	2	3	4	5	6	7	8	9	10	11	12
CO No.	CO1	CO1	CO2	CO2	CO2	CO1	CO1	CO1	CO2	CO2	CO1	CO2
Bloom's Level* (L1 to L6)	L1	L2	L3	L2	L1	L3	L3	L3	L2	L3	L3	L2
Weightage CO1: 26.5						Weightage CO2: 23.5						

*Bloom's Level: L1: Remember, L2: Understand, L3: Apply, L4: Analyze, L5: Evaluate, L6: Create

Note: Answer **all** the sections with all the questions

Section-A

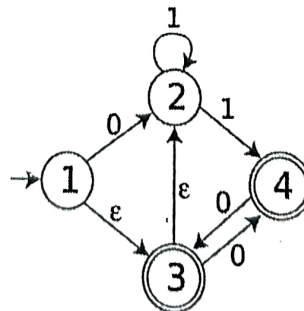
(2*5=10)

1. Define the term "Alphabet" in the context of automata theory.
2. Differentiate between DFA and NFA.
3. Construct Deterministic Finite Automata (DFA) to accept string that always ends with 101 over alphabet $\Sigma = \{0,1\}$.
4. State Arden's Theorem.
5. Explain in brief about the Kleen's Theorem.

Section-B

(5*5=25)

6. Construct a Finite automata (DFA) which accepts all binary numbers whose decimal equivalent is divisible by 4 over $\Sigma = \{0, 1\}$.
7. Construct the given NFA with ϵ -transition to DFA.



8. Differentiate between Mealy and Moore Machine. Construct a Moore machine equivalent to the Mealy machine given in the following table:

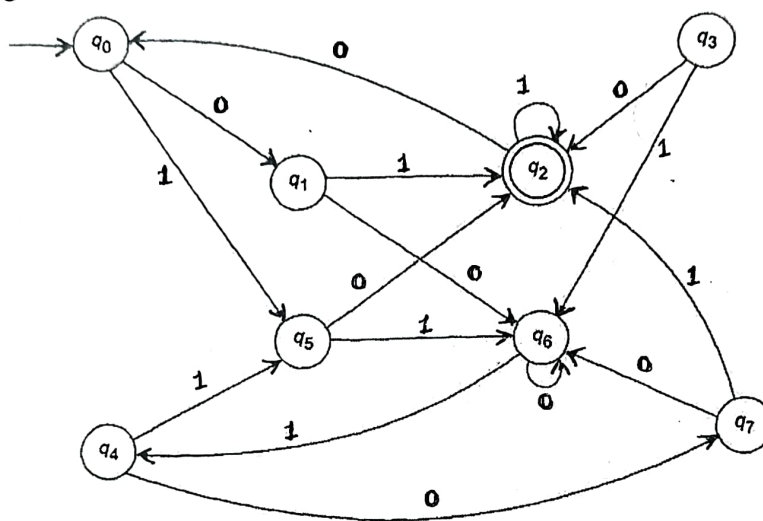
Present States	Input: a		Input: b	
	state	O/P	State	O/P
q1	q1	1	q2	0
q2	q4	1	q4	1
q3	q2	1	q3	1
q4	q3	0	q1	1

9. Construct finite automata equivalent to following regular expression $(1+0)^*(11+00)(1+0)^*$.
10. Find a regular expression corresponding to the following language over the alphabet $\{a, b\}$:
- The set of all strings that contain exactly two a's.
 - The set of all strings containing the substring aa.
 - The set of all strings that begin and end with either a or b.

Section-C

(7.5*2=15)

11. Minimize the given DFA shown below.



12. Compute the regular expression using Arden's Theorem for the following DFA. States: q_0 (Initial), q_0, q_1 (Final); Transition: $\delta(q_0, 0) \rightarrow q_1$, $\delta(q_0, 1) \rightarrow q_2$, $\delta(q_1, 0) \rightarrow q_1$, $\delta(q_1, 1) \rightarrow q_2$, $\delta(q_2, 0) \rightarrow q_0$, $\delta(q_2, 1) \rightarrow q_2$.

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